

115.109

Introduction to Business Analytics

Topic 5: Panels Sampling

Dr Andrew Murphy

(Please contact me via Communication Tools in Stream)

Week (starting)	Business Problem	Data & Analysis	Chapter (Black et al)	Assessments
1 (13/7)	Understanding Data: discounts & margin	Calculating data in Excel; GST, discounts and Margins	1	Lab1
2 (20/7)	Operational Monitoring: Dashboards	Tabulating & Graphing	2	Lab2
3 (27/7)	Industry trends	Statistics NZ: retail survey Descriptive stats: averages	3	Lab3
4 (3/8)	Customer trends	Statistics NZ: census Descriptive stats: variation	3	Lab4
5 (10/8)	Transaction data: product sales	Transaction data sources. Sampling	7	Lab5
6 (17/8)	Loyalty cards	Loyalty card schemes. Probability; proportions, χ^2	4, 12	Lab6 Test 1
Mid	Semester	Study	Period	
7 (7/9)	Segmentation	Statistics NZ: census. Normal distribution; Means	6, 8	Lab7
8 (14/9)	Determining difference (1)	Means; ANOVA	9, 10, 11	Lab8
9 (21/9)	Associating and Predicting	Explaining relationships: Correlation & regression	13	Lab9
10 (28/9)	Determining Difference (2)	Explaining relationships (continued)	14	-
11 (5/10)	Trends over time	Time series, Indices Unstructured data: social media	15	Lab11 Test 2
12 (12/10)	Corporate reporting	Finance Dashboards reporting		Report

2026	Assessment	Lab 1&2	Lab 3&4	Lab 5&6	Test 1	Lab 7&8	Lab 9,11	Test 2	Report
Week	13-Jul	O: 13.07							
commencing	20-Jul		O: 20.07						
Monday:	27-Jul								
	03-Aug	D: 03.08							
	10-Aug			O: 10.08					
	17-Aug		D: 17.08		O: 14.08 C: 21.08				O 17.08
	<i>Mid-Semester Study Period</i>			D:31.08					
	07-Sep					O: 07.09			
	14-Sep								
	21-Sep						O: 21.09		
	28-Sep								
	05-Oct					D: 05.10		O: 02.10 C: 09.10	
	12-Oct						D: 19.10		D: 19.10
		O=opens (available from); D=due (without late penalty); C=closes (not available after)							

Group Information Survey

Contents

- ↓ [Assessment Overview](#)
- ↓ [Assessment Planner](#)
- ↓ [Formal Requirements to pass this course](#)
- ↓ [Assessment resources](#)
- ↓ [Assessment 1 information: Computer Exercises](#)
- ↓ [Missing AI Declarations?](#)
- ↓ [Labs 1 & 2](#)
- ↓ [Labs 3 & 4](#)
- ↓ [Assessment 2 & 3 information: Online Tests](#)
- ↓ [Optional Reviews](#)
- ↓ [Assessment 4 information](#)
- ↓ [Analytical Report](#)
- ↓ [Report Groups](#)



QUESTIONNAIRE

Established group identifier: closes 10 August

26 groups
so far

Enter the Student IDs of ALL of the group members (either 2 or 3 members per group) IF you have already formed a group, or know of other students in the class you would like to form a group with.

If you have not yet formed a group, I will allocate you - complete the [Group Information Survey](#) to assist this.

53 as of 9/8



QUESTIONNAIRE

Group Information Survey: closes 10 August

If you have not already formed a group, please complete this survey by Monday August 10

This survey will help me generate teams of similar learners for Assignment 4 (group report). All information in the survey is confidential and is used solely for the purpose of organising teams for this assessment.

Teams created using the survey information tend to be better organised than teams randomly assigned.

Items with black * in the question will be shared publicly with your teammates (name, email and city only). The red asterisk next to the question number indicates a required question. This information is not shared.

Distribution Transformations:

	B	C	D	E	F	G
	5 ink cartridges priced at \$25, 30, 35, 40 and \$45					
3 Xi		25	30	35	40	45

Concept 1: *adding* a constant to a set of numbers:

$\bar{X}$ increases by that constant, but s is unchanged

Concept 2: *multiplying* a set of numbers by a constant:

$\bar{X}$ and s are both multiplied by that constant

5 ink cartridges priced at \$25, 30, 35, 40 and \$45 (ex-GST)

1. *What is the mean and standard deviation for these cartridges?*

A) 175, 35; B) 35, 7.91; C) 5, 20; D) 35, 35

2. *What happens to the mean and std. dev. once GST of 15% is added?*

A) Both increase by 15%; B) Both decrease by 15%; C) Nothing;

D) The mean increases by 15% but std. dev. remains the same

3. *What happens to the mean and standard deviation once a delivery charge of \$2 per cartridge is added (to the inclusive of GST price)?*

A) Both increase by \$2; B) Both decrease by \$2; C) Nothing;

D) The mean increases by \$2 but std. dev. remains the same

More on politics... UK Election

https://en.wikipedia.org/wiki/Opinion_polling_for_the_next_United_Kingdom_general_election

[hide] Dates conducted	Pollster	Client	Area	Sample size	Con	Lab	Lib Dems	SNP	Green	Reform	Others	Lead
4 Jul 2024	2024 general election	—	UK	—	23.7%	33.7%	12.2%	2.5%	6.8%	14.3%	6.8%	10.0
			GB		24.4%	34.7%	12.5%	2.5%	6.9%	14.7%	4.3%	10.3
3 Jul	Number Cruncher Politics	N/A	GB	2,496	23%	41%	11%	2%	7%	13%	[show] 3%	18
2–3 Jul	JL Partners	<i>The Rest Is Politics</i>	GB	2,005	23%	38%	13%	3%	5%	17%	[show] 3%	15
2–3 Jul	Savanta	N/A	UK	2,101	20%	39%	10%	2%	5%	17%	7%	19
2–3 Jul	We Think	N/A	GB	1,210	23%	41%	11%	2%	7%	15%	2%	18
1–3 Jul	Survation	<i>Good Morning Britain</i>	GB	1,679	20%	38%	12%	3%	7%	17%	[show] 3%	18
1–3 Jul	Norstat	N/A	GB	3,134	24%	37%	11%	4%	6%	16%	[show] 2%	13
1–3 Jul	Opinium	N/A	GB	2,219	21%	41%	11%	2%	7%	17%	[show] 1%	20
1–3 Jul	Ipsos	N/A	GB	2,076	19%	37%	11%	5%	9%	15%	[show] 5%	18
29 Jun – 3 Jul	Deltapoll	N/A	GB	1,737	22%	39%	10%	2%	7%	17%	[show] 3%	17
2 Jul	PeoplePolling	GB News	GB	1,260	16%	36%	10%	4%	9%	20%	5%	16
1–2 Jul	Whitestone Insight	<i>Daily Express/Daily Mirror</i>	GB	2,008	21%	38%	10%	3%	7%	18%	[show] 3%	17
30 Jun – 2 Jul	BMG	<i>The i</i>	GB	1,854	22%	39%	11%	3%	7%	16%	[show] 3%	17
28 Jun – 2 Jul	Techne	N/A	GB	5,503	21%	40%	11%	3%	6%	16%	3%	19
28 Jun – 2 Jul	Redfield & Wilton	N/A	GB	20,000	22%	41%	10%	3%	6%	16%	[show] 2%	19
26 Jun – 2 Jul	Survation	<i>Good Morning Britain</i>	UK	1,022	18%	38%	11%	3%	7%	17%	[show] 6%	20
19 Jun – 2 Jul	YouGov (MRP)	N/A	GB	47,751	22%	39%	12%	3%	7%	15%	[show] 3%	17



Sampling

- Population $\Rightarrow$ sample frame $\Rightarrow$ sample size $\Rightarrow$ sampling method
- Sampling method and size vary with the project
 - cost, time, acceptable _____, subgroup analysis
- Probability sampling
 - simple random sample
 - _____ sampling
 - stratified random sample
 - multistage sampling
- Non-probability sampling
 - strict quota sample
 - volunteer sample
 - “theoretically informed judgement”
 - _____ sample
 - snowball sample

Populations and Samples

Source: Gray & Guppy (1994): 138, 145

■ Who is the target group for the study?	→	This is called the study population
■ Who <i>in</i> the target group should be surveyed?	→	This is called the sample
■ How <i>many</i> people should be surveyed?	→	This is called the sample size
■ How should the people to be surveyed be <i>selected</i> ?	→	This is called the sampling method

If 70% of a sample "agrees," then with a sample size of	the population value might be	
	as low as	as high as
10	34%	94%
50	55%	83%
100	59%	78%
250	64%	76%
1000	67%	73%

Why Sample?

- A *sample* is some part of the population (i.e. a 'subset' of the population) that is actually available as a source of information

It is not always practical to analyse the entire data relating to a particular problem. This may be due to:

1. Not physically possible to collect the entire data / reach the population
2. Too expensive to collect the entire data
3. The data may be _____ more quickly by using a sample rather than the complete set

Sample Bias

- A sample should be *unbiased*
 - That is, there should be no tendency for certain individuals to have a larger or smaller chance of being selected in the sample
- It should be genuinely _____ of the population
- If there is any systematic *bias* the result will be valueless
- Bias may be generated even if the sample selected is sufficiently representative
 - E.g. in interviewing situations, where the wording of the questions and behaviour or appearance of the interviewer can combine to distort the outcome of sample surveys

Sample Bias

There are several simple rules that can be followed to help eliminate bias in sampling. These include:

- Do *not* use only people who _____ to be in the sample
- Do *not* choose a sample using a method that _____ of the population (eg only landlines)
- Do *not* use people in the sample only because they are 'handy' or readily available (“_____”)
- Ensure that the person selecting the sample does not have a _____ in the results of the sample

Even so it is very difficult to eliminate bias completely—the best we can hope for is to limit it as much as possible

Sampling Error

- Even if sampling is completely free of bias, there is still fluctuation due to naturally occurring random variation
- It is necessary to assess how much _____ can be expected to occur from one sample to another
- This random discrepancy between a measurement from a sample (called a _____) and the population quantity being estimated (the *population parameter*) is called *sampling error*
- A _____ (using the sampling error) can be constructed to determine the range of possible values for the population parameter

continued

Sample Size

- Large samples (in general) yield more accurate estimates, but they cost more in time and money
- Doubling the size of a sample does *not* double the _____
- If a sample is biased, _____ the sample size only increases the bias
- Random samples are usually more _____ than non-random ones, since all members of the population must be _____ and interviewers must sometimes travel long distances to ensure that all groups are _____
- *How* should the members be selected for inclusion in the sample?

Sampling Method: Random

- Selecting numbers from a table of random numbers as published in books or generated by a computer
- In their simplest form, these tables consist of a series of the digits 0 to 9 that have (as far as can be ascertained) an equal chance of occurring
- How to select within the table?
- Random number generator in Excel: `=rand()`
- Generates a proportion (between 0 and 1)
- To get a specific range: multiply by the maximum e.g. whole number between 1 and 30: `=round(rand()*30,1)`
- To get a large number: multiply by 'a large number e.g. `round(rand()*100000,1)`

Sampling Method: Systematic

- In *systematic sampling*, the members of the population are put in the form of an *array*
- That is, they are arranged in some natural order (usually ascending or descending)
- Then every n th item is selected for the sample
- The value of n should be chosen so that it will yield a sufficiently large sample
- The greatest problem seems to arise when the listing of the population is incomplete
E.g. taking every tenth number out of the telephone directory may give a reasonable selection of the people listed there, but it discriminates strongly against people who do not have a telephone and those who have 'silent' numbers

Sampling Method: 'Sophisticated'

- quota sampling
 - stratified sampling
 - cluster sampling
 - multistage sampling
 - importance sampling
-
- Gathering representative data from a group, within strict specification (100 females, 20 children aged 10-14...) Representative individuals are systematically chosen out of a specific subgroup.

For example, a researcher might ask for a sample of 100 females, or 100 individuals between the ages of 20-30.
 - Used when a specific sub-sample size is required (but since this is non-random, statistical comparisons are problematic)

Sampling Method: 'Sophisticated'

- quota sampling
 - stratified sampling
 - cluster sampling
 - multistage sampling
 - importance sampling
-
- Divides the population into strata (subsets), then takes samples of approximately equal size from each stratum
 - Each stratum is homogeneous within (eg the same household income class) and heterogeneous between
 - The selection of subjects from each stratum is done randomly
 - The individual is the sampling unit
 - May cost more to obtain than random sample but the precision of results is usually greater

-

<https://keydifferences.com/difference-between-stratified-and-cluster-sampling.html>

- THE ENGINE
OF THE NEW
NEW ZEALAND

Sampling Method: 'Sophisticated'

- quota sampling
 - stratified sampling
 - cluster sampling
 - multistage sampling
 - importance sampling
-
- Multistage is an extension of cluster sampling: the population is first broken down into a set of distinct groups, from which a number of groups are selected randomly
 - The groups selected are again broken down according to another characteristic, and a random selection is again made

Sampling Method: 'Sophisticated'

- quota sampling
 - stratified sampling
 - cluster sampling
 - multistage sampling
 - importance sampling
- This is where the probability of selecting a particular individual (or larger subset) is dependent on its importance
- For example, in election predictions, there is often little purpose in sampling electorates where the result is a 'foregone conclusion'
 - If the 'swinging' electorates can be identified, the efforts of the pollsters can be concentrated on effective sampling from these

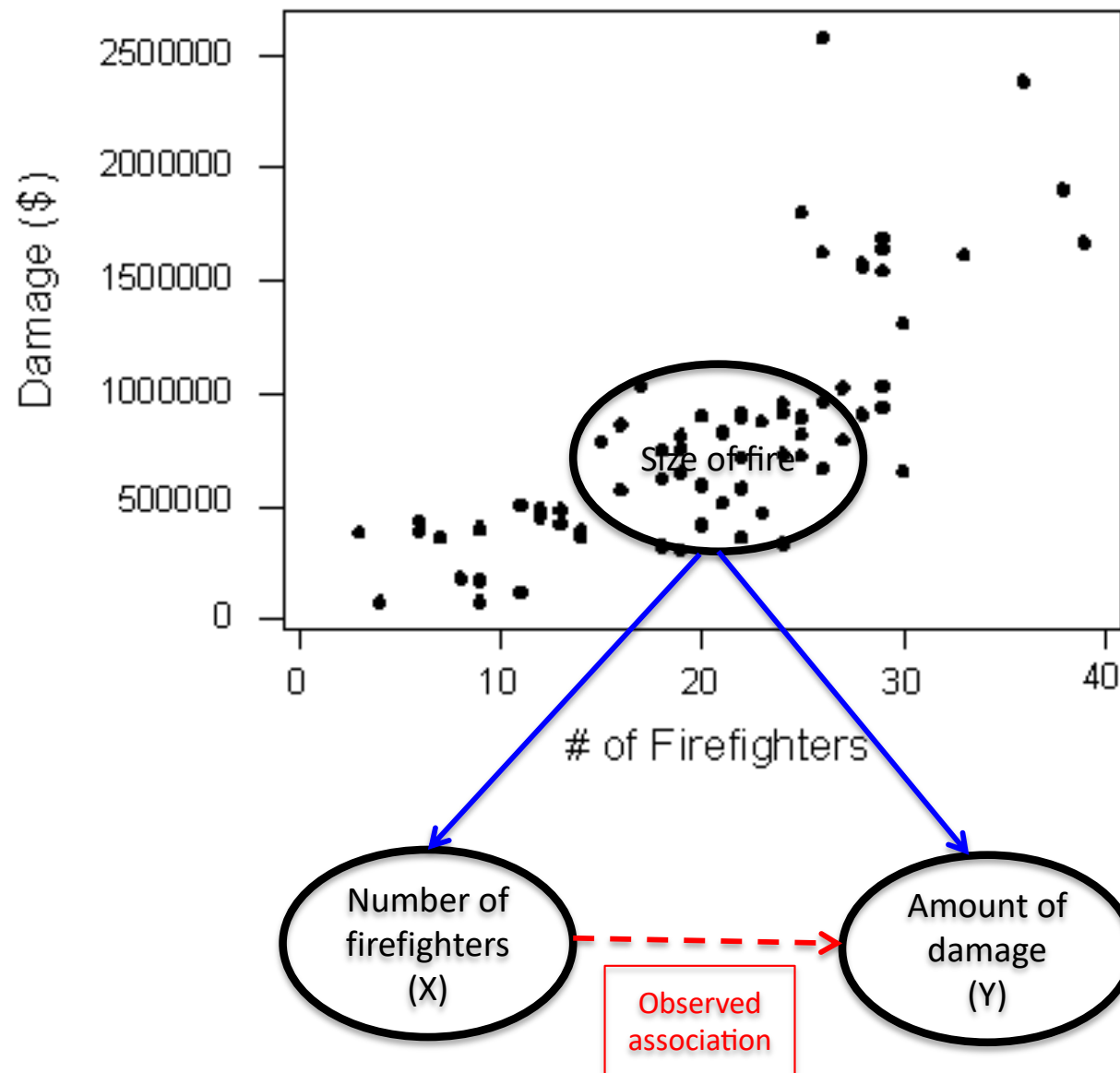
Types of Questions & Data

- Classification
 - classifies people, their circumstances and environment
 - e.g. age, income, housing tenure, household size etc
- Behaviour
 - relates to the behaviour of people
 - e.g. where do they shop? what sort of things do they buy?
- Attitudes
 - relates to peoples' attitudes, opinions and beliefs
 - typically the most difficult to collect due to potential for patterned responses, insincerity and response errors

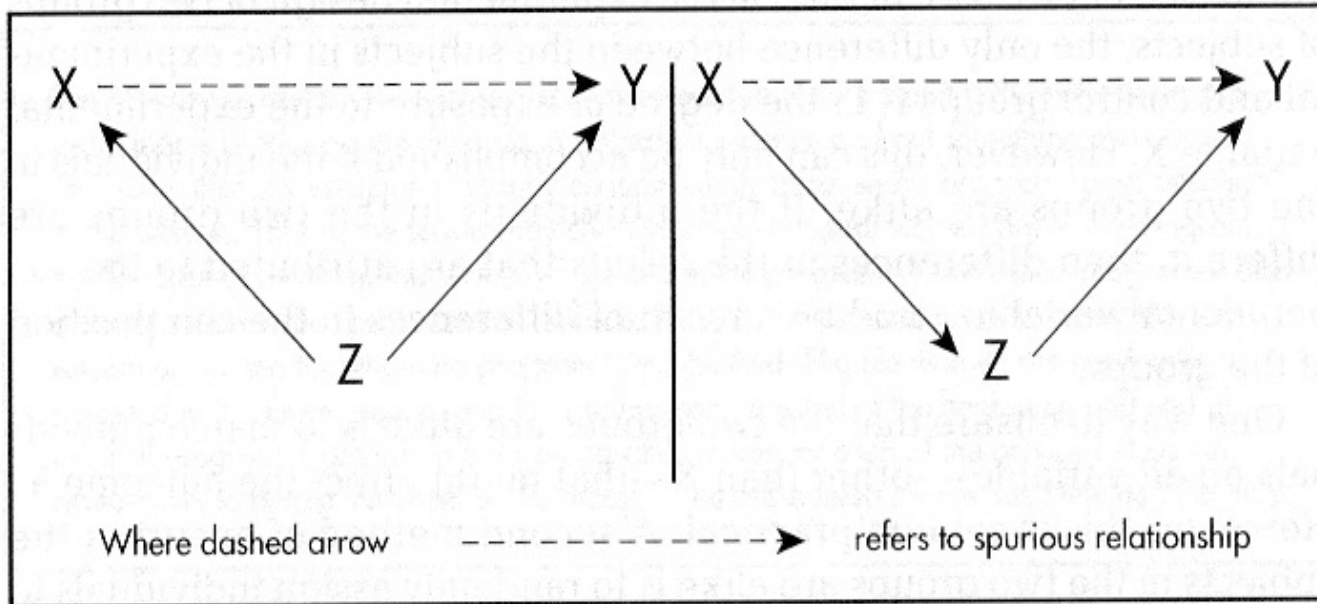
Purpose of Surveys

- A broad distinction:
 - *descriptive surveys*: observing patterns in a population e.g. market research and opinion polls
 - *analytic surveys*: addressing ‘why’ questions, identifying relationships between variables, producing data for statistical analysis
- Key types of survey variable
 - dependent variables (variables for which explanation is being sought)
 - experimental or independent variables (all possible predictors)
 - controlled and non-controlled variables

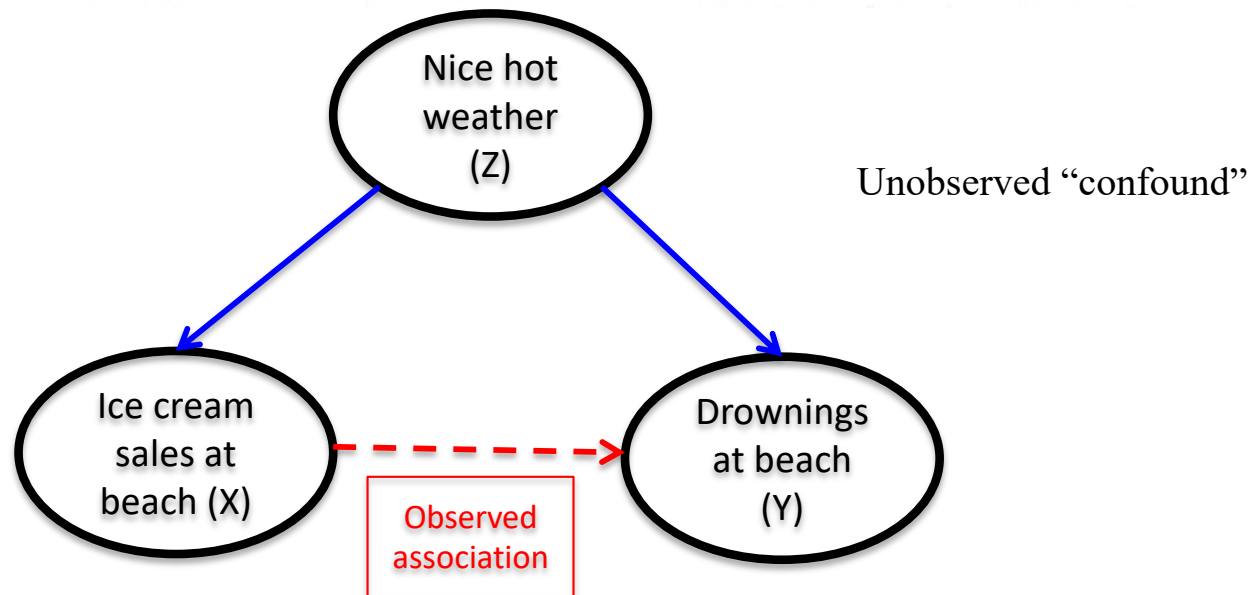
Relationships and



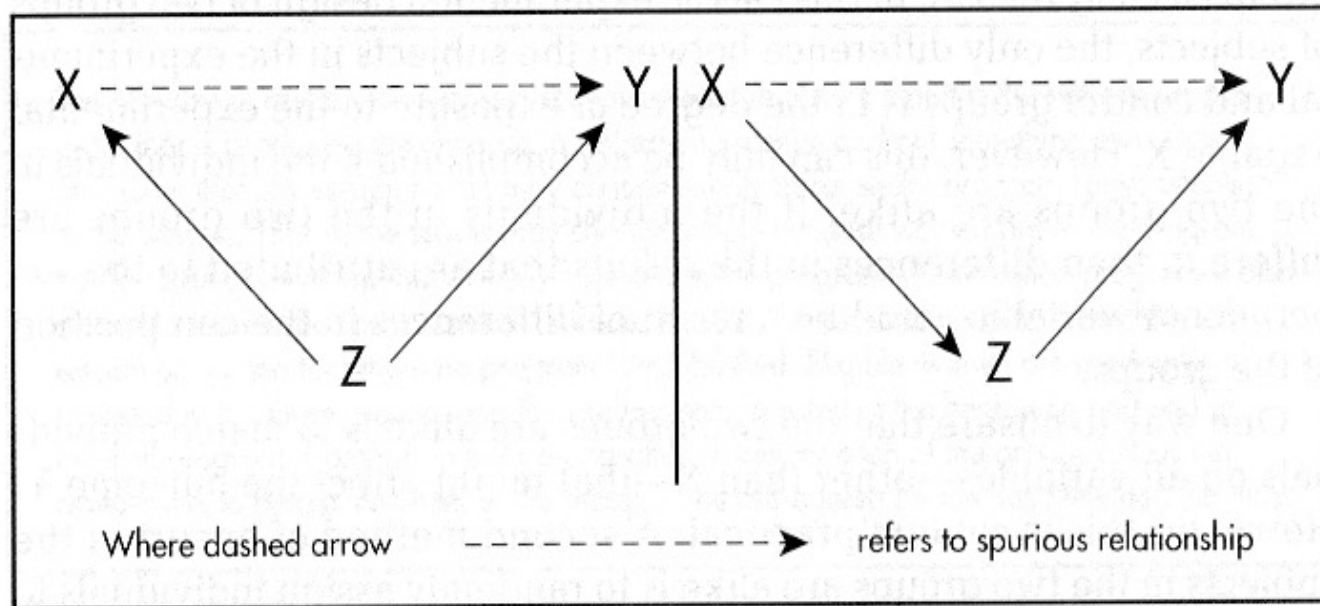
Variables: _____ Relationships



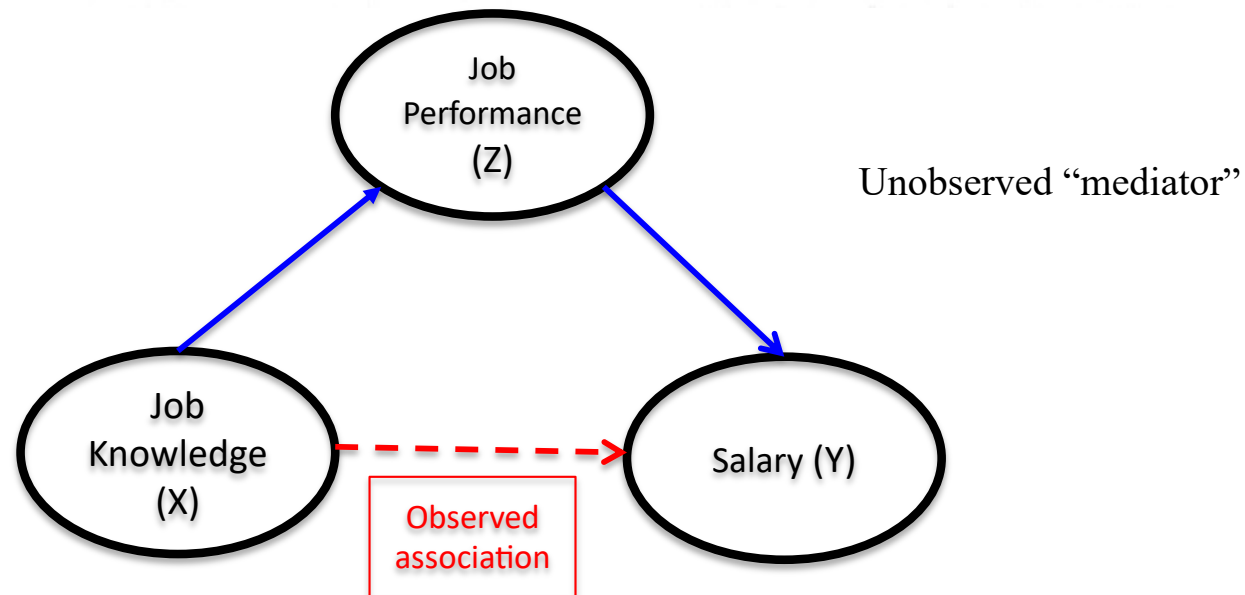
Source: Gray & Guppy
(1994): 47



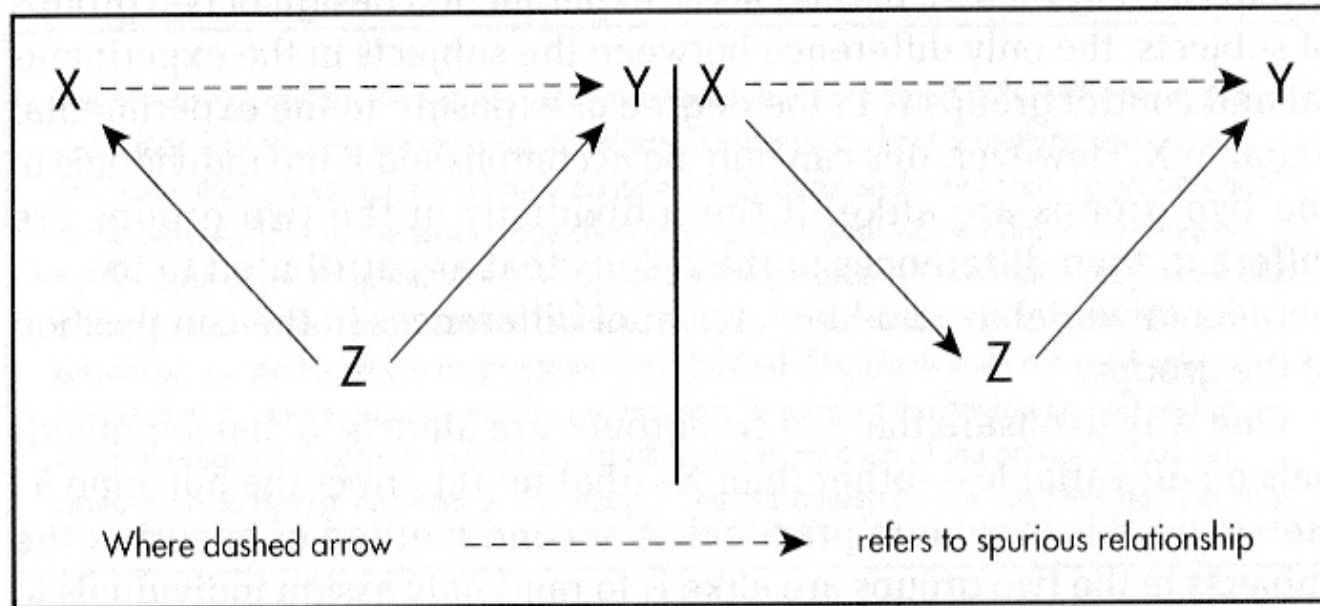
Variables: _____



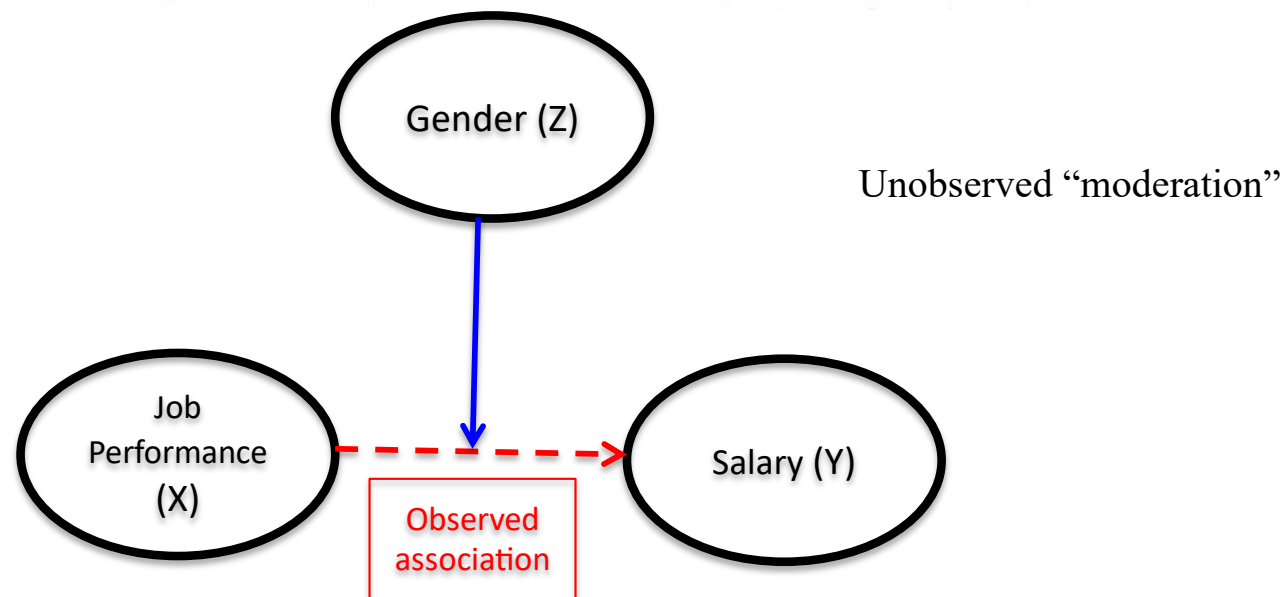
Source: Gray & Guppy
(1994): 47

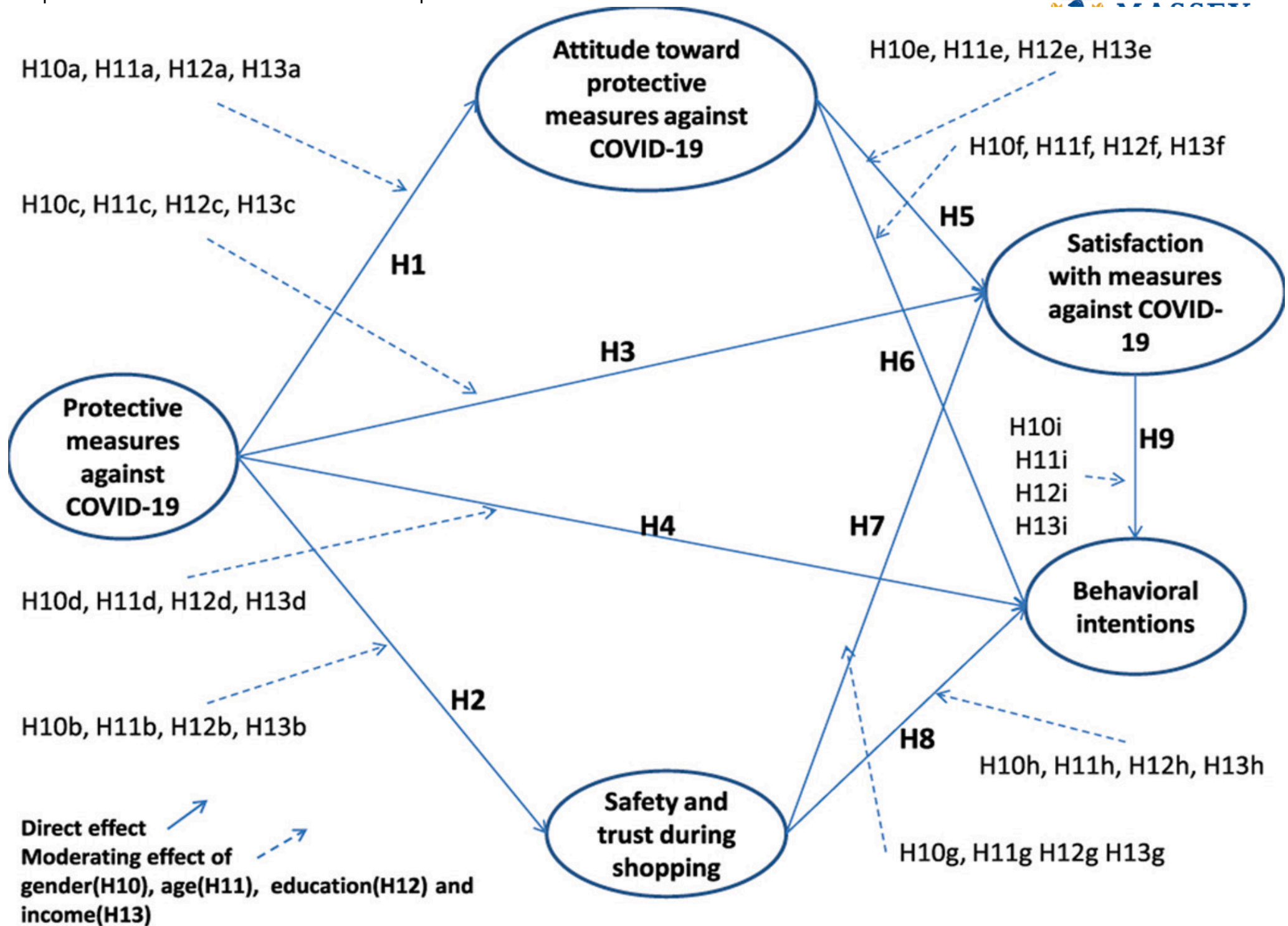


Variables: _____



Source: Gray & Guppy
(1994): 47





Format & Progression

- Introduction/welcome (identification and anonymity)
- Progression (general – detailed/sensitive – general)
- Enhancing flow: block similar questions and use continuity statements
- Using filters to save time (if Q7 = no, go to ...)
- Appropriate level of complexity: avoid jargon

Designing Questions

- Question styles:
 - close-ended/structured (easier to ask, answer and analyse)
 - open-ended/unstructured (“in own words”)
 - discrete vs continuous (quantitative continuum)
- Things to avoid:
 - lack of focus
 - leading questions (introduce bias)
 - double-barrelled questions (avoid “and” and “or”)
 - negative and double-negative questions (avoid “not”)
 - vague time-span

Designing Questions

- Asking sensitive classificatory questions
 - age, income, profession
- Measuring Perception
 - assume a relation between attitude and behaviour (problematic)
 - Likert five point scales
 - Attitude Choice statements
 - Semantic Differential Scales
- The importance of a pilot study

Design: Likert-type

To what extent do you agree with the following statements:

https://masseybusiness.asia.qualtrics.com/jfe/form/SV_b2bpGACbVnQQnNX

	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree
I am willing to spend more money in order to buy products that are sustainable	☐	☐	☐	☐	☐
	1	2	3	4	5
I believe it is acceptable to spend extra money for products that are sustainable	☐	☐	☐	☐	☐
Retailers should be required to take back packaging they sell product in (including cardboard, bubble wrap or filler, polystyrene...)	☐	☐	☐	☐	☐
There should be a bottle deposit scheme, with customers reimbursed for taking back milk, juice and water bottles to the retailer or a recycle station	☐	☐	☐	☐	☐

Design: Attitude Choice

These are some general questions about **what you think** about waste.

Auckland Council Peoples Panel:
Rubbish & Recycling survey

How much do you **agree or disagree** that:

	1 - Strongly disagree	2	3	4	5 - Strongly agree	I don't know
It is important for Auckland to reduce the amount of rubbish going to landfill	☐	☐	☐	☐	☐	☐

Individual actions are important to reduce the amount of rubbish going to landfill	☐	☐	☐	☐	☐	☐
--	-----------------------	-----------------------	-----------------------	-----------------------	-----------------------	-----------------------

Why did you give those ratings?

If you can't think of anything please leave this question blank.

Design: Semantic Differential

Question: *On a scale of 1 to 7, please tell me how you feel about the proposed wind farm for the Wiggly Green site. Is it ...*

(use SHOW CARD and read out items) ...

beautiful	☐ ☒ ☐ ☐ ☐ ☐ ☐	ugly
noisy	☐ ☐ ☐ ☐ ☒ ☐ ☐	quiet
advanced technology	☐ ☐ ☒ ☐ ☐ ☐ ☐	backward technology
unsafe	☐ ☐ ☐ ☐ ☐ ☐ ☒	safe
reliable	☐ ☐ ☐ ☐ ☐ ☒ ☐	unreliable

Source: Flowerdew & Martin (1997): 92

Design: Layout

First Name of Household Member	Relationship to Respondent *	Birthdate (Month-Year)	Gender	Presently Employed? **
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____

* Write in appropriate category: spouse, child, parent, sibling, other relative, friend, boarder, other (specify).

** Refers to paid work. Include part-time work.

Source: Gray & Guppy (1994): 108, 112

SECTION E **COMPUTERS** The following questions address the issues of computerization of your company/firm as well as the use of computers by the members of your association.

27. Does your company/firm have computer facilities? (Please check the appropriate answer)

☐ No ☐ Yes

28. What types of computer facilities does your company/firm have? (Please check as many categories as apply)

☐ Microcomputers (stand-alone)
☐ Microcomputer network
☐ Microcomputers (tied to mainframe)
☐ Terminals tied to mainframe
☐ Mainframe (access limited to specialists)

29. The applications **software** for your company/firm's computer(s) was: (Please check as many categories as apply)

☐ Fully custom-designed
☐ Packaged software with major modifications
☐ Packaged software with few or no modifications

30. Have you used a computer? (Please check the appropriate answer)

☐ No ☐ Yes

31. What types of computers have you used? (Please check as many categories as apply)

☐ Microcomputers ☐ Mainframe

32. Do you currently use a microcomputer?

☐ No ☐ Yes

33. Where do you use a microcomputer? (Please check as many as apply)

☐ at home ☐ at the office ☐ other _____
 (please specify)

34. What are the brand names of the computers you use: (Please write in as many brand names as apply)

at home _____
 at office _____
 other _____



Lab 5: Sample Comparison

115.109: Topic 5 Lab Worksheet: Sustainability Sampling @ 10%

Due date: 31 August 2026

Version: 1 (6 August 2026)

Your Name:

Student ID:

Task 1: Compare Demographic profiles of the 10% and Full datasets

Process: Repeat Task 1 from Lab 4, on the original (reduced) SustainabilityData-10%sample sheet

1) Copy and Paste Special > Paste Special > Values and **Transpose** the counts you found for Gender in Lab 4 task 1.1 in the table below, column C

Repeat the Copy > paste Special for Age (Lab 4 task 1.2), and then Income (Lab 4 task 1.4), Paste Special > Values and Transpose each time

2) Repeat the steps for Lab 4 Task 1 steps 1, 2 & 4 on the 10% sample data (see SustainabilityData-10% sample tab)

Simplest method: create a PivotTable (Insert > Pivot table, in a new sheet, which you modify for each variable) of Gender, Age and Income, and copy > Paste Special > Formula the results here

Hint: Because there are now so many more rows, you will definitely want to use the keyboard shortcuts to navigate around: Control/Command + ⬇ (down arrow key) scrolls to the bottom of a group of data, stopping at the last filled cell. Likewise Cntrl/Cmd + ⬆ to go up to the top of a group of data; Cntrl/Cmd + ⬇ to go right and Cntrl/Cmd + ⬅ to go left. Holding down the Shift key as well selects the range of cells as you go.

Demographic Values	Frequency Counts		% of Respondents	
	Full Dataset	10% Sample	Full Dataset	10% Sample
Male				
Female				
Gender Diverse				
<18				
18-24				
25-34				
35-44				
45-54				
55-64				
65+				
1: Up to \$30,000				
2: \$30,001 - \$50,000				
3: \$50,001 - \$70,000				
4: \$70,001 - \$100,000				
5: \$100,001-\$130,000				
6: Over \$130,000				
7: Rather not say				
# Respondents	1042	104		

Rationale:

This lab returns to the reduced SustainabilityData-10% sample from Labs 1 and 2, and compares that dataset to the dataset used in Labs 3&4. The full dataset (SustainabilityData) comprises 1042 rows and 90 data columns, whereas the Sustainability-10%sample file used in Labs 1&2 comprised 104 rows and 33 data columns, as a 10% sample of the full dataset.

This sampling was done using the function =rand() in each row, which creates a random decimal number between 0 and 1. I copied the random numbers and Paste Special > Values to 'fix' them (otherwise Excel recalculates the random numbers each time you do something), and sorted smallest to largest to randomly select 104 rows. I also deleted some questions (columns) to simplify the sheet a bit.

For this lab we will contrast the 10% sample with the full dataset.

Tasks 1-2 repeat Lab 1's Task 4 and Lab 2's Task 2 for the full dataset, and to compare with what you found in the earlier labs.

Task 3 asks you to analyse the full sample, filtering the dataset to compare some demographics segments.

Task 4 asks you to use the PivotTable feature to do this more quickly.

3) Complete the % of respondents columns above, then interpret the table: what patterns do you see (if any)?

<<Table Interpretation>>

Lab 5: Sample comparison

3) Complete the % of respondents columns above, then interpret the table: what patterns do you see (if any)?

<<Table Interpretation>>

Task 2: Contrast Sustainability knowledge between 10% and Full datasets

1) Copy Lab 2 task 2 result

	Frequency Counts				% of Respondents			
	Column AI & CK: "I have a good knowledge about sustainability issues"		Column AK & CM: "I know a lot about sustainable products and brands sold in the marketplace"		Sustainability Issues Knowledge		Sustainability Brands Knowledge	
	Full Dataset (CK)	10% Sample (AI)	Full Dataset (CM)	10% Sample (AK)	Full Dataset (CK)	10% Sample (AI)	Full Dataset (CM)	10% Sample (AK)
Response level								
1								
2								
3								
4								
5								
# Respondents	1042	104	1042	104				

2) Interpret the table: what patterns do you see (if any)?

<<Interpretation>>

Lab 5: Filtering

Murphy, Andrew:

In case these task statements don't format correctly (eg are truncated*), they are reproduced to the right
* for some unknown reason Excel seems to get confused with the graphic elements - at least on my Mac

Task 3: Analyse customer segments, using the Filter tool

Filtering is a powerful tool in quickly reducing a dataset to what you are interested in seeing. There are some tricks in using it efficiently.

We will use the Age variable in column C to split the data, by combining ("recoding") age categories (1+2, 3+4, 5+6)

1) In the SustainabilityData sheet, click on cell C1053 and enter the CountIf formula twice, for age category 1 + 2

Repeat in cells C1054 for age categories 3+4, and C1055 for age categories 5+6.

The recoded age categories should have pretty similar sample sizes. (334-362), since Perceptive made efforts to have a balanced sample

Excel offers a more flexible method for selectively analysing data: Filtering

2) In the SustainabilityData sheet, click on cell C4, and holding the Shift and Control/Command key, press the down arrow. This should select cells C4:C1046

3) Now select Filter from the AZ Sort & Filter button (Home ribbon). This adds a down arrow to each variable. Click on the down arrow within the Age column then click the tick/check marks next to the age categories you don't want (3,4,5,6,7) to **un**-select them (leaving 1+2 ticked, and some other stuff below)

The result will be a reduced set of rows showing: those that meet the Age criteria. All others are now hidden.

The usual CountIf function unfortunately doesn't work with filtering: it counts **all** the rows, even the filtered-out (hidden) ones.

Check it! Have the Age category counts in cells C1053:C1055 changed? The solution is to use the **Subtotal** function instead.

4) In cell C1056, insert **=subtotal(2,C5:C1046)**. {The Subtotal command can take many forms, and COUNT is option 2.}

{Some other useful options: 1: Average; 4: Max; 5: Min; 7: StDev; 9: Sum. Try them! But return to 2 for count}

The easy way to switch between subtotals of the segments is simply to change the filtering segment.

But if you want to **compare** the segments, you need to capture the information each filter generates.

Hopefully the rows beneath row 1046 have not been hidden when you turned on the filter (sometimes Excel might hide rows we need).

5) In cell CE1048 insert the formula **=SUBTOTAL(2,CE5:CE1046)** to calculate the sample sizes, and copy the result across to CM1048

6) In cell CE1049 insert the formula **=SUBTOTAL(1,CE5:CE1046)** to calculate the mean, and copy the result across to CM1049

7) Copy the range CE1049:CM1049, and Paste Special > Values in cell CE1052. Also copy cell CE1048, and Paste Special > Values into cell CN1052

8) Change the filter in SustainabilityData cell C4 from 1&2 (ages 18-34) to 3 and 4 (ages 35-54). When you click outside the Filter box (or close it), the selected rows will change

Copy the means and Paste Special > Values into CE1053, and then copy CE1048 and Paste Special > Values into CN1053

9) Change the filter in cell C4 from 3&4 (35-54) to 5&6 (55+) and copy and Paste Special > Values all the means into CE1054, and copy CE1048 and Paste Special > Values into CN1054

10) Remove the filter in cell C4 (choose Select All) and ensure that cells CE1052:CN1054 are filled with the means for the three age groups, and the sample sizes.

11) With the filter removed, also copy and Paste as Values the means for all ages combined into cells CE1051:CM1051, and the Count into CN1051

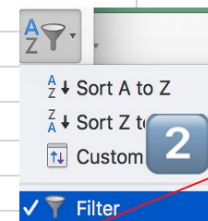
12) In SustainabilityData, copy CE1051:CM1054, and Paste Special > Paste Special > Values and Transpose into cell C89 of Lab 5 sheet (following table):

13) Also transfer the Counts from cells CN1051 to CN1054 into cells C98:F98

Bonus Task:

For the very keen: also calculate standard deviations **=subtotal(7,CE5:CE1046)**
duplicate the three age s and place beside the means results

Recoded Age Category: Means



Lab 5: Filtering

Task 3: Analyse customer segments, using the Filter tool

Filtering is a powerful tool in quickly reducing a dataset to what you are interested in seeing. There are some tricks in using it efficiently.

We will use the variable "Country of Sample" in column C to split the data, by combining ("recoding") age categories (1+2, 3+4, 5+6)

1) In the SustainabilityData sheet, click on cell C1053 and enter the CountIf formula twice, for age category 1 + 2

Repeat in cells C1054 for age categories 3+4, and C1055 for age categories 5+6.

The recoded age categories should have pretty similar sample sizes. (334-362), since Perceptive made efforts to have a balanced sample

Excel offers a more flexible method for selectively analysing data: Filtering

2) In the SustainabilityData sheet, click on cell C4, and holding the Shift and Control/Command key, press the down arrow. This should select cells C4:C1046

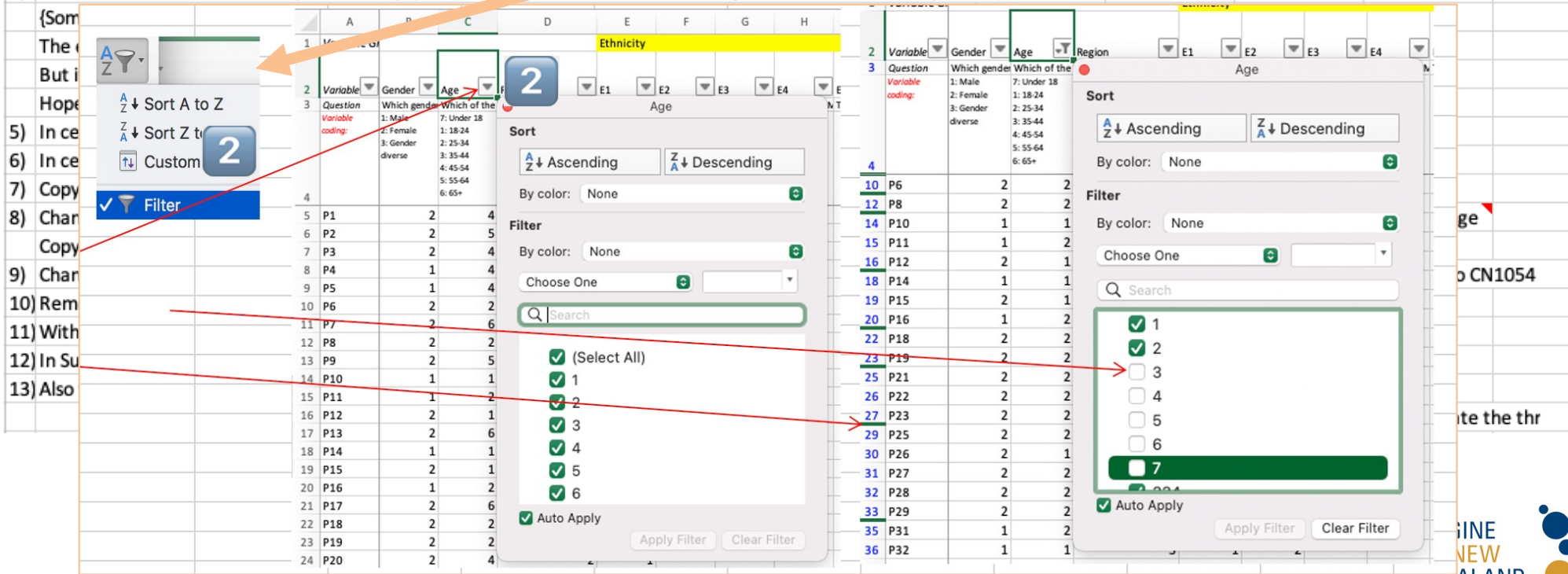
3) Now select Filter from the AZ Sort & Filter button (Home ribbon). This adds a down arrow to each variable. Click on the down arrow within the Age column then click the tick/check marks next to the age categories you don't want (3,4,5,6,7) to un-select them (leaving 1+2 ticked, and some other stuff below)

The result will be a reduced set of rows showing: those that meet the Age criteria. All others are now hidden.

The usual CountIf function unfortunately doesn't work with filtering: it counts all the rows, even the filtered-out (hidden) ones.

Check it! Have the Age category counts in cells C1053:C1055 changed? The solution is to use the Subtotal function instead.

4) In cell C1056, insert =subtotal(2,C5:C1046). {The Subtotal command can take many forms, and COUNT is option 2.}



The screenshot displays the Excel 'Filter' task pane. In the 'Filter' section, the 'Age' variable is selected, and a list of age categories (1, 2, 3, 4, 5, 6, 7) is shown with checkboxes. Categories 1 and 2 are checked, while 3, 4, 5, 6, and 7 are unchecked. The 'Auto Apply' checkbox is also checked. The 'Sort' section shows 'Ascending' selected. The background shows the 'SustainabilityData' sheet with columns A through H. Red arrows point from the text instructions to the corresponding parts of the Excel interface.

Lab 5: Filtering

- 7) Copy the range CE1049:CM1049, and Paste Special > Values in cell CE1052. Also copy cell CE1048, and Paste Special > Values into cell CN1052
- 8) Change the filter in SustainabilityData cell C4 from 1&2 (ages 18-34) to 3 and 4 (ages 35-54). When you click outside the Filter box (or close it), the selected rows will change
Copy the means and Paste Special > Values into CE1053, and then copy CE1048 and Paste Special > Values into CN1053
- 9) Change the filter in cell C4 from 3&4 (35-54) to 5&6 (55+) and copy and Paste Special > Values all the means into CE1054, and copy CD1048 and Paste Special > Values into CN1054
- 10) Remove the filter in cell C4 (choose Select All) and ensure that cells CE1052:CN1054 are filled with the means for the three age groups, and the sample sizes.
- 11) With the filter removed, also copy and Paste as Values the means for all ages combined into cells CE1051:CM1051, and the Count into CN1051
- 12) In SustainabilityData, copy CE1051:CM1054, and Paste Special > Paste Special > Values and Transpose into cell C89 of Lab 5 sheet (following table):
- 13) Also transfer the Counts from cells CN1051 to CN1054 into cells C98:F98

Bonus Task:

For the very keen: also calculate standard deviations =subtotal(7,CE5:CE1046)
duplicate the three ages and place beside the means results

			Recorded Age Category: Means		
Column	Product	All Ages	18-34	35-54	55+
CE	Capable to solve				
CF	Contribute				
CG	Can help				
CH	Purchase effect				
CI	Positive effect				
CJ	Individual difference				
CK	Good knowledge				
CL	Know sustainability				
CM	Know sustainable brands				
	# Respondents				

5) Comment on the pattern(s) you see

Choose the data that you want

☒ Select a table or range

Table/Range:

☐ Use an external data sou

Lab 5: Pivoting

Task 4: Analyse customer segments using the PivotTable tool

A quicker (but not necessarily easier) way to complete Task 3 is to use PivotTables

1) In cell B4 of the SustainabilityData sheet, click Insert > Create PivotTable in a new worksheet, changing the default data range to \$B\$4:\$CM\$1046

2) A new sheet is created with a PivotTable template with many selectable fields (to the right-hand side of the screen)

If you click off the white dialog box (or a table it creates), the PivotTable Fields dialog box will disappear. Just click on this box/table to make it reappear.

PivotTable7

Click in this area to work with the PivotTable report



PivotTable Fields

FIELD NAME Search fields

- ☐ Gender
- ☒ Age
- ☐ Region
- ☐ E1
- ☐ E2

PivotTable Fields

FIELD NAME Search fields

- ☐ Ethical products
- ☐ Household Income
- ☒ Capable to solve
- ☒ Contribute
- ☒ Can help
- ☒ Purchase effect

Filters

Columns

Values

Rows

Age

Values

Sum of Contrib...

PivotTable Fields

FIELD NAME Search fields

Sum of Contrib...

Sum of Capable to solve

Summarize by Show data as

Sum

Count

Average

Row Labels Average of Capable to solve Average of Contribute Average of Can help

1	3.3340757	3.7550111	3.5345212
2	3.3925550	3.9390863	3.6582064
3	2.5000000	3.5000000	3.0000000
(blank)			
Grand Total			

EXAMPLE: by Gender, not Age

↓

Row Labels	'Capable to solve	'Can Contribute	'Can help	Sample size
Male	3.3	3.8	3.5	449
Female	3.4	3.9	3.7	591
Gender Diverse	2.5	3.5	3.0	2
Total sample	3.4	3.9	3.6	1042

Insert PivotTable by Age classes here: